

GANGHUA WANG

Email: wang9019@umn.edu

Website: <https://gwang.umn.edu/>

Address: 492 Ford Hall, 224 Church Street SE
Minneapolis, MN 55455

EDUCATION

University of Minnesota, Twin Cities

Ph.D in Statistics

Aug. 2019 - present

Minneapolis, MN

Peking University

B.S. in Statistics, Minor in Economics

Sept. 2015 - July 2019

Beijing, China

EXPERIENCE

Research assistant: Deep learning theory

Supervised by Prof. Jie Ding, University of Minnesota

Sept. 2021 - Present

Minneapolis, MN

- Topics have not been decided yet.

Research assistant: Model privacy

Supervised by Prof. Jie Ding, University of Minnesota

Jan. 2021 - Aug. 2021

Minneapolis, MN

- We formalize the definition of model privacy and propose model defense mechanisms that provide rigorous guarantees.

Research assistant: Data privacy

Supervised by Prof. Jie Ding, University of Minnesota

June 2020 - Aug. 2020

Minneapolis, MN

- We proposed a novel framework to protect data privacy for each individual, also known as local data privacy. Instead of adding noise to provide uncertainty, we report a subset contains true value.

Project: Travelers Modeling Problem

Peace of Mind Insurance Company

Nov. 2019 - Dec. 2019

Minneapolis, MN

- A competition to predict future policies based on the historical insurance data, held in [Kaggle](#). Collaborated with other graduate students, we ranked first in the public leaderboard, and second in the private leaderboard.

Summer internship: Component analysis in brain image

Supervised by Prof. Hongyu Zhao, Yale University

July. 2018 - Sept. 2018

New Haven, CT

- With ADHD fMRI datasets, we proposed to extract functional brain neural network components using Latent Dirichlet Allocation method, which facilitates the understanding of the intrinsic structure of human brain and the diagnosis of ADHD.

Research assistant: Community detection with co-variates

Supervised by Prof. Jinzhu Jia, Peking University

Dec. 2017 - June 2018

Beijing, China

- We adopted a maximum likelihood estimation based method to combine the information from both co-variates and the network to detect communities under the stochastic block model.

PUBLICATION

Submitted

* indicates equal contributions

† indicates corresponding author(s) (if not the senior author)

boldface indicates myself

- [1] Gen Li, **Ganghua Wang**, Jie Ding[†], and Yutao Gu. “Provable Identifiability of ReLU Neural Networks via LASSO Regularization”. In: *arXiv preprint* (2021).
- [2] **Ganghua Wang** and Jie Ding[†]. “Subset Privacy: Draw from an Obfuscated Urn”. In: *arXiv preprint arXiv:2107.02013* (2021).

Manuscript

- [3] **Ganghua Wang**, Jie Ding[†], and Yuhong Yang. “Model privacy”. In: *arXiv preprint* (2021).
- [4] **Ganghua Wang**, Jie Ding[†], and Yuhong Yang. “Regression with Set-Valued Categorical Predictors”. In: *arXiv preprint* (2021).

HONORS AND AWARDS

Awarded by University of Minnesota, Twin Cities

Summer Research Fellowship *2020*

School of Statistics First Year Scholarship *2019*

Awarded by Peking University

The Academic Excellence Scholarship (3 times) *2016 - 2018*

Fang Zheng Scholarship *2017*

Wu Si Scholarship *2016, 2018*

Freshman Scholarship *2015*

TEACHING

Teaching Assistant, University of Minnesota, Twin Cities

STAT 4102 Theory of Statistics II *Fall 2020*

STAT 3021 Introduction to Probability and Statistics *Spring 2020*

STAT 3011 Introduction to Statistical Analysis *Fall 2019*

TECHNICAL STRENGTHS

Computer Languages Python, MATLAB, R, SQL